

Navjot Singh

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EDUCATION

University of Illinois at Urbana-Champaign

Ph.D. in Computer Science *Advisor: Edgar Solomonik*

Urbana, IL

May 2020 – Aug 2025

University of Illinois at Urbana-Champaign

M.S. in Applied Mathematics *Advisor: Edgar Solomonik*

Urbana, IL

Aug 2018 – May 2020

Delhi Technological University

B.Tech. in Mathematics and Computing

New Delhi, India

Aug 2014 – May 2018

EXPERIENCE

Postdoctoral Scholar Advisor: Xiaoye Sherry Li

Lawrence Berkeley National Laboratory

Sep 2025 –

Berkeley, CA

- Working on efficient numerical methods for learning PDEs with Gaussian Processes
- Developed a provably accurate approximation to the gradient of Lanczos method for bilinear matrix functions of Hermitian matrices

Research Intern Advisor: Ramakrishnan Kannan

Oak Ridge National Laboratory

June 2024 - Aug 2024

Oak Ridge, TN

- Developed scalable optimization algorithms for generalized Tucker tensor completion for knowledge graph completion.
- Our algorithms outperform state-of-the-art optimization algorithms in terms of accuracy and time to solution on distributed-memory architecture.

Research Intern Advisor: Xiaoye Sherry Li, Yang Liu

Lawrence Berkeley National Laboratory

May 2023 - Aug 2023

Berkeley, CA

- Designed a tensor-network completion algorithm for highly oscillatory operators achieving 4-digit accuracy with $O(N \log N)$ storage, outperforming existing methods by orders of magnitude.

Research Intern Advisor: Chao Yang

Lawrence Berkeley National Laboratory

May 2022 - Aug 2022

Berkeley, CA

- Built tensor-decomposition and floating-point compression pipeline for eigenvector storage in many-body eigenvalue problems, achieving $10\times$ compression with only 1 additional LOBPCG iteration.
- Developed a novel neural network architecture and optimization method for representing nonlinear functionals in combination with CP tensor decomposition.

Graduate Research Assistant

University of Illinois at Urbana-Champaign

May 2019 - Aug 2025

Urbana, IL

- Developed parallel numerical algorithms for compression and reconstruction of high dimensional data
- Developed efficient sparse tensor computational kernels for implementing algorithms for reconstruction problems in distributed-memory setting

TECHNICAL SKILLS

Languages: Python, C++

HPC & Parallel: MPI, OpenMP, distributed-memory algorithms, Cyclops Tensor Framework

Scientific Computing: NumPy, SciPy, PyTorch, numerical linear algebra, iterative/direct solvers, tensor computations

Methods: numerical optimization, randomized algorithms, Riemannian optimization, low-rank/sparse methods

PUBLICATIONS

1. **Navjot Singh**, Edgar Solomonik, Xiaoye Sherry Li, and Yang Liu. Efficient tensor completion algorithms for highly oscillatory operators. *Numerical Linear Algebra with Applications*, 33(4):e70098, 2026
2. **Navjot Singh** and Edgar Solomonik. Alternating Mahalanobis distance minimization for accurate and well-conditioned CP decomposition. *SIAM Journal on Scientific Computing*, 45(6):A2781–A2812, 2023
3. **Navjot Singh**, Zecheng Zhang, Xiaoxiao Wu, Naijing Zhang, Siyuan Zhang, and Edgar Solomonik. Distributed-memory tensor completion for generalized loss functions in Python using new sparse tensor kernels. *Journal of Parallel and Distributed Computing*, 169:269–285, 2022
4. **Navjot Singh**, Linjian Ma, Hongru Yang, and Edgar Solomonik. Comparison of accuracy and scalability of Gauss–Newton and Alternating Least Squares for CANDECOM/PARAFAC decomposition. *SIAM Journal on Scientific Computing*, 43(4):C290–C311, 2021
5. Chaoqi Yang, Cheng Qian, **Navjot Singh**, Cao (Danica) Xiao, M Westover, Edgar Solomonik, and Jimeng Sun. Atd: Augmenting cp tensor decomposition by self supervision. In *Advances in Neural Information Processing Systems*, volume 35, pages 32039–32052, 2022
6. Chaoqi Yang, **Navjot Singh**, Cao Xiao, Cheng Qian, Edgar Solomonik, and Jimeng Sun. Mtc: Multiresolution tensor completion from partial and coarse observations. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, KDD ’21, page 1953–1963, New York, NY, USA, 2021. Association for Computing Machinery

IN REVIEW

1. **Navjot Singh**, Kipton Barros, and Xiaoye Sherry Li. Fast and stable gradient approximation for bilinear forms of hermitian matrix functions. *arXiv preprint arXiv:2605.12801*, 2026

AWARDS

Kenichi Miura Award for Outstanding Accomplishment in High Performance Computing (500\$)
Graduate College Presentation Award 2022: SIAM PP’22 (Virtual registration: 150\$)
SIAM Student Travel Award: SIAM MDS’22 (Domestic, ~\$900), SIAM LA’24 (International, ~\$1100)
Travel Award Housholder Symposium XXII: Travel support for attending the Householder symposium (\$2325)
SIAM Early Career Travel Award: SIAM PP’26 (International, \$1550)

ACADEMIC SERVICE

Reviewer: TMLR, ACM TOPC, IEEE TPDS, Springer FoCM, Elsevier AMMOD, ACM TOMS, Springer AAM, Elsevier PARCO, SISC, KDD’26, ICPP’26, NeurIPS’26

SELECTED INVITED PRESENTATIONS

SIAM CT’25 (Montréal, Canada): Large-Scale Tensor Decomposition and Completion for Non-Gaussian Data
SIAM LA’24 (Paris, France): Riemannian Optimization on Low Rank Tensor Manifolds
ICIAM’23 (Tokyo, Japan): Efficient Algorithms and Software for Generalized Tensor Completion
SIAM CSE’23 (Amsterdam, Netherlands): Distributed-Memory Algorithms for Tucker Tensor Completion